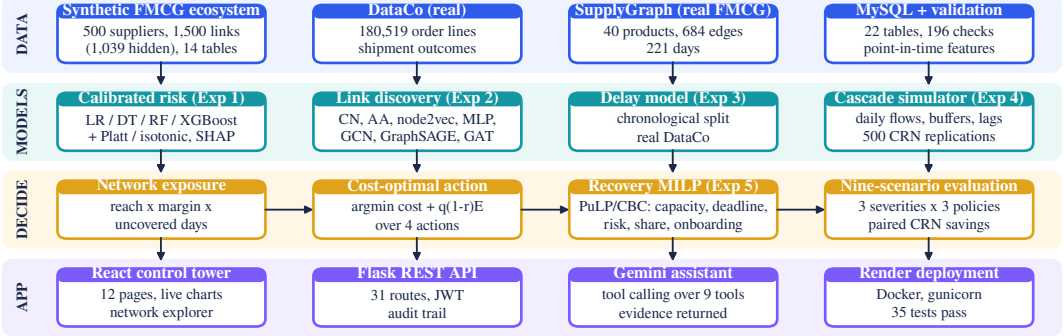
DATA
Synthetic FMCG ecosystem
500 suppliers, 1,500 links
(1,039 hidden), 14 tables
DataCo (real)
180,519 order lines
shipment outcomes
SupplyGraph (real FMCG)
40 products, 684 edges
221 days
MySQL + validation
22 tables, 196 checks
point-in-time features
MODELS
Calibrated risk (Exp 1)
LR / DT / RF / XGBoost
+ Platt / isotonic, SHAP
Link discovery (Exp 2)
CN, AA, node2vec, MLP,
GCN, GraphSAGE, GAT
Delay model (Exp 3)
chronological split
real DataCo
Cascade simulator (Exp 4)
daily flows, buffers, lags
500 CRN replications
DECIDE
Network exposure
reach x margin x
uncovered days
Cost-optimal action
argmin cost + q(1-r)E
over 4 actions
Recovery MILP (Exp 5)
PuLP/CBC: capacity, deadline,
risk, share, onboarding
Nine-scenario evaluation
3 severities x 3 policies
paired CRN savings
APP
React control tower
12 pages, live charts
network explorer
Flask REST API
31 routes, JWT
audit trail
Gemini assistant
tool calling over 9 tools
evidence returned
Render deployment
Docker, gunicorn
35 tests pass